Polyhedral Combinatorics

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Chapter 1

Preliminaries

1.1 Vector and affine spaces

Vector notation

Throughout the text, let $V = \mathbb{R}^n$ denote a real vector space of finite dimension n. Let V^* denote the dual vector space of V, that is, the vector space of all linear functionals $V \to \mathbb{R}$. Elements of V will commonly be named x, while elements of V^* will commonly be named y. The symbol "0" will be used to denote both the scalar 0 and vector 0.

For most of this text we will work without coordinates. In case we wish to use coordinates, we fix a *standard basis* $e_1, \ldots, e_n$ of V. This determines a *standard dual basis* $e^1, \ldots, e^n$ of V^* defined by $e^i(e_j) = 0$ if $i \neq j$ and $e^i(e_i) = 1$ for all i.

It is common practice to always fix an inner product $\langle \cdot, \cdot \rangle$ and identify V and V^* using $x \mapsto \langle x, \cdot \rangle$. In this text I have decided to keep V and V^* formally distinct.

We denote the linear span of a set $S \subset V$ by $\operatorname{span} S$. Given two sets $S \subset V$, $T \subset W$ in vector spaces V, W, we say that S and T are *linearly equivalent* if there is a linear isomorphism $f : \operatorname{span} S \to \operatorname{span} T$ such that $f(S) = T$.

Given a vector subspace $W \subset V$, we define the *orthogonal complement* or *annihilator* of W to be the following vector subspace of V^*:

$$W^\perp = \{y \in V^* : y(x) = 0 \text{ for all } x \in W\}.$$

Affine spaces

An *affine subspace* of V is a set of the form $x + W$, where W is a linear subspace of V and $x + W$ denotes the translation of W by a vector $x \in V$. The *dimension* of $x + W$ is the dimension of W. In this text, the term *affine space* will refer to an affine subspace, although we may not explicitly name the ambient vector space.

An *affine combination* of points $x_1, \ldots, x_m \in V$ is any point of the form $\sum_{i=1}^m \lambda_i x_i$ where $\lambda_1, \ldots, \lambda_m$ are real numbers such that $\sum_{i=1}^m \lambda_i = 1$. If A is an affine subspace of V, then any affine combination of elements of A is also in A. The *affine span* of a nonempty set $S \subset V$ is the smallest affine subspace of V containing S. Equivalently, it is the set of all affine combinations of elements of S. We denote the affine span of S by $\operatorname{aff} S$. We define the *dimension* of S to be the dimension of its affine span. We say a set is *full-dimensional* if its affine span is V.

The points $x_1, \ldots, x_m \in V$ are *affinely independent* if for any $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$ such that $\sum_{i=1}^m \lambda_i x_i = 0$ and $\sum_{i=1}^m \lambda_i = 0$, we have $\lambda_1 = \cdots = \lambda_m = 0$. For points $x_1, \ldots, x_m \in V$, the following are equivalent.

(i) $x_1, \ldots, x_m$ are affinely independent.
(ii) $x_1 - x_m, x_2 - x_m, \ldots, x_{m-1} - x_m$ are linearly independent.
(iii) $(x_1, 1), \ldots, (x_m, 1)$ are linearly independent in $V \times \mathbb{R}$.

It follows that the maximum size of an affinely independent set in V is $\dim V + 1$.

Let A, B be affine spaces (with possibly different ambient vector spaces). An *affine map* is a function $f : A \to B$ such that for any $x_1, \ldots, x_m \in A$ and affine combination $\sum_{i=1}^m \lambda_i x_i$, we have $f(\sum_{i=1}^m \lambda_i x_i) = \sum_{i=1}^m \lambda_i f(x_i)$. An *affine isomorphism* is an affine map which is a bijection. Given two subsets S, T of (possibly different) vector spaces, we say that S and T are *affinely equivalent* if there is an affine isomorphism $f : \operatorname{aff} S \to \operatorname{aff} T$ such that $f(S) = T$.

Given any set $S \subset V$, we define $S^\perp \subset V^*$ to be the orthogonal complement of the linear subspace parallel to $\operatorname{aff} S$. In other words, $S^\perp$ is the set of all linear functionals in V^* which are constant on S.

1.2 Polyhedra

A *convex polyhedron* in V is a set of the form

$$\{x \in V : y_i(x) \leq b_i \text{ for all } i \in I\} \tag{1.2.1}$$

where I is a finite set and $y_i \in V^*$, $b_i \in \mathbb{R}$ for all $i \in I$. We call a representation of a convex polyhedron in the form (1.2.1) a *constraint presentation* of the polyhedron. If there is no risk of confusion, we may abbreviate the above presentation as

$$\{y_i(x) \leq b_i,\ i \in I\}.$$

The inequalities $y_i(x) \leq b_i$ are called *linear constraints*, or *constraints*. When writing a presentation, we may write some constraints as $y_i(x) \geq b_i$ (equivalent to $(-y_i)(x) \leq -b_i$) or $y_i(x) = b_i$ (equivalent to the combined constraints $y_i(x) \leq b_i$ and $y_i(x) \geq b_i$).

Every polyhedron has many presentations. For example, if $y_1(x) \leq b_1$ and $y_2(x) \leq b_2$ are constraints in a presentation, we may add to the presentation any nonnegative linear combination of these constraints (i.e., $(c_1y_1 + c_2y_2)(x) \leq c_1b_1 + c_2b_2$ for $c_1,\ c_2 \geq 0$) without changing the polyhedron. However, we will see in Theorem 3.2.5 that there is an essentially unique minimal way to present a polyhedron.

One can equivalently define convex polyhedra in terms of affine spaces and affine functionals (Exercise 1.1). In particular, this makes it clear that the property of being a polyhedron is preserved under affine isomorphism.

For the rest of this text, we will refer to convex polyhedra as simply *polyhedra*. In other contexts, the term "polyhedron" refers more generally to a finite union of convex polyhedra.

It is clear from the definition that all polyhedra are closed sets, and the intersection of a finite number of polyhedra is a polyhedron. A polyhedron which is bounded is called a *polytope*.

A *cone* is a subset $C \subset V$ satisfying the following conditions: $0 \in C$, and if $x \in C$, then $\lambda x \in C$ for all real $\lambda > 0$. A *polyhedral cone* is a polyhedron which is a cone. We note the following fact (Exercise 1.2).

Lemma 1.2.2. If C is a cone and $y \in V^*$, then either $\max_C y = 0$ or y is unbounded from above on C.

It follows from the lemma that a subset of V is a polyhedral cone if and only if it is of the form

$$\{y_i(x) \leq 0,\ i \in I\}$$

for some finite set I and $y_i \in V^*$ for all $i \in I$.

Example 1.2.3 (Simplicial cones)**.** Recall that $e^1,\dots,e^n$ is the standard dual basis of V^*. The polyhedral cone

$$\mathbb{R}^n_{\geq 0} := \{x \in \mathbb{R}^n : e^i(x) \geq 0,\ i = 1, \dots, n\}$$

is the *nonnegative orthant* of $\mathbb{R}^n$. Any cone linearly equivalent to this cone is called a *simplicial cone*.

Example 1.2.4 (Half-spaces)**.** Given any $y \in V^*$ and $b \in \mathbb{R}$, the set

$$\{y(x) \leq b\}$$

defined by a single constraint is a polyhedron. If $y \neq 0$, this polyhedron is a *(closed) half-space* of V. If $y = 0$, then the polyhedron is empty if $b < 0$ and the whole space V if $b \geq 0$.

Example 1.2.5 (Affine subspaces)**.** Every affine subspace of V can be written as

$$\{y_i(x) = b_i,\ i \in I\}$$

for some finite set I and $y_i \in V^*$, $b_i \in \mathbb{R}$. Hence every affine subspace is a polyhedron. An affine subspace of the form $\{y(x) = b\}$ is an *affine hyperplane*, or just *hyperplane*. A *linear hyperplane* is an affine hyperplane which contains the origin.

Example 1.2.6 (Cubes)**.** The polyhedron

$$Q = \{x \in \mathbb{R}^n : -1 \leq e^i(x) \leq 1,\ i = 1, \dots, n\}$$

is an n-dimensional polytope called a *cube* or *n-cube*.

Example 1.2.7 (Cross-polytopes)**.** The polyhedron

$$R = \{x \in \mathbb{R}^n : \pm e^1(x) \pm \cdots \pm e^n(x) \leq 1\},$$

where the constraints go through all 2^n combinations of $+$ or $-$, is an n-dimensional polytope called a *cross polytope*.

Example 1.2.8 (Simplices)**.** The polyhedron

$$\Delta = \{x \in \mathbb{R}^{n+1} : e^1(x) + \cdots + e^{n+1}(x) = 1,\ e^i(x) \geq 0,\ i = 1, \ldots, n\}$$

is an n-dimensional polytope called a *simplex* or *n-simplex*. Note that we have defined this polytope in $\mathbb{R}^{n+1}$, but it has an n-dimensional affine span, specifically $\{x \in \mathbb{R}^{n+1} : e^1(x) + \cdots + e^{n+1}(x) = 1\}$. In general, an *$n$-simplex* is defined to be any polytope affinely equivalent to Δ.

Another example of an n-simplex is

$$\{x \in \mathbb{R}^n : e^1(x) + \cdots + e^{n+1}(x) \leq 1,\ e^i(x) \geq 0,\ i = 1, \ldots, n\}$$

which is full-dimensional in $\mathbb{R}^n$.

You're probably also familiar with the regular *dodecahedron* and *icosahedron*, which, along with the regular 3-simplex, 3-cube, and 3-cross polytope, form the Platonic solids in $\mathbb{R}^3$.

Exercises

1.1. Let A be an affine space. An *affine functional* on A is an affine map $f : A \to \mathbb{R}$. A subset of A is a convex polyhedron if and only if it is of the form

$$\{x \in A : y_i(x) \leq 0 \text{ for all } i \in I\}$$

where I is a finite set and each y_i is an affine functional on A.

1.2. If C is a cone and $y \in V^*$, then either $\max_C y = 0$ or y is unbounded from above on C.

Chapter 2

Convex sets

2.1 Convex sets and cones

A set $K \subset V$ is *convex* if for any $x_1,\ x_2 \in K$, the line segment with endpoints x_1, x_2 is contained in K; in other words, if $(1-\lambda)x_1 + \lambda x_2 \in K$ for all $0 < \lambda < 1$.

It is easy to check that all polyhedra are convex (hence the name "convex polyhedron"). In fact, any set of the form

$$\{x \in V : y_i(x) \leq b_i,\ i \in I\}, \tag{2.1.1}$$

where I is a (possibly infinite) set and $y_i \in V^*$, $b_i \in \mathbb{R}$ for all $i \in I$, is a closed convex set. We call a representation of the form (2.1.1) a *constraint presentation* of a convex set. We will show in Corollary 2.2.5 that all closed convex sets have a constraint presentation. Polyhedra are precisely the sets which have a finite constraint presentation.

Convex hull

A *convex combination* of points $x_1,\ \ldots,\ x_m \in V$ is any point of the form $\sum_{i=1}^m \lambda_i x_i$ where $\lambda_1,\ \ldots,\ \lambda_m$ are nonnegative real numbers satisfying $\sum_{i=1}^m \lambda_i = 1$. In other words, a convex combination is an affine combination in which the weights are restricted to be in $[0,1]$. If K is convex, then any convex combination of elements of K is also in K.

Let $S \subset V$. The *convex hull* of S, denoted $\operatorname{conv} S$, is the smallest convex subset of V which contains S. This is well-defined because the intersection of all convex sets which contain S is again a convex set. Equivalently, the convex hull of S is the set of all convex combinations of elements of S.

Example 2.1.2. The cube in Example 1.2.6 is the convex hull of the 2^n points $\pm e_1 \pm \cdots \pm e_n$. The cross-polytope in Example 1.2.7 is the convex hull of the $2n$ points $\pm e_1, \ldots, \pm e_n$.

Example 2.1.3. The simplex Δ in Example 1.2.8 is the convex hull of the $n+1$ points $e_1, \ldots, e_{n+1}$. In general, a set is an n-simplex if and only if it is the convex hull of $n+1$ affinely independent points.

Convex cones

A *conical combination* of vectors $x_1, \ldots, x_m \in V$ is any vector of the form $\sum_{i=1}^m \lambda_i x_i$ where $\lambda_1, \ldots, \lambda_m \geq 0$. A conical combination of an empty set of vectors is defined to be 0. If C is a convex cone, then every conical combination of elements of C is also in C.

Let $S \subset V$. The *conical hull* of S, denoted $\operatorname{cone} S$, is the smallest convex cone which contains S. The conical hull of S equals the set of all conical combinations (including 0) of elements of S.

Example 2.1.4. The nonnegative orthant in Example 1.2.3 is the conical hull of the n vectors $e_1, \ldots, e_n$. In general, a set is an n-dimensional simplicial cone if and only if it is the conical hull of n linearly independent vectors.

Convex cones are often more convenient to work with than general convex sets. We can use the following trick to move between the two settings. Given a set $S \subset V$, we can embed it in the vector space $V \times \mathbb{R}$ as $S \times \{1\}$. Then

$$\operatorname{conv} S = \{x \in V : (x, 1) \in \operatorname{cone}(S \times \{1\})\}. \tag{2.1.5}$$

Carathéodory's Theorem

As previously discussed, every point in the convex hull (resp. conical hull) of a set S can be written as a convex combination (resp. conical combination) of finitely many points in S. Carathéodory's Theorem says that in fact only a bounded number of points are needed.

Theorem 2.1.6 (Carathéodory's Theorem)**.** Let S be a subset of $V = \mathbb{R}^n$.

(a) If $x \in \operatorname{conv} S$, then x is a convex combination of $n+1$ or fewer elements of S.
(b) If $x \in \operatorname{cone} S$, then x is a conical combination of n or fewer elements of S.

Proof. Part (a) can be proved by applying part (b) to $S \times \{1\}$ and using (2.1.5). Thus, it suffices to prove (b).

Suppose that $x \in \operatorname{cone} S$. Then we can write $x = \sum_{i=1}^{m} \lambda_i x_i$ for some $x_1, \ldots, x_m \in S$ and nonnegative $\lambda_1, \ldots, \lambda_m$. If $m \leq n$, we are done. Otherwise, suppose $m > n$. Since V has dimension n, the vectors $x_1, \ldots, x_m$ are linearly dependent. Let $\mu_1, \ldots, \mu_m$ be real numbers such that $\sum_{i=1}^{m} \mu_i x_i = 0$ and the μ_i are not all 0. Then for any real number α, we have

$$x = \sum_{i=1}^{m} \lambda_i x_i - \alpha \sum_{i=1}^{m} \mu_i x_i = \sum_{i=1}^{m} (\lambda_i - \alpha \mu_i) x_i.$$

By replacing every μ_i with $-\mu_i$ if necessary, we may assume at least one of the μ_i is positive. Start with $\alpha = 0$ and move α towards ∞. At some point, at least one of the $\lambda_i - \alpha\mu_i$ will equal 0, and the other $\lambda_i - \alpha\mu_i$ will be nonnegative. Hence, we can write x as a conical combination of $m - 1$ elements of S. Repeat this procedure until x is written as a conical combination of n elements of S. $\square$

We will use this theorem to prove the following.

Proposition 2.1.7. The convex hull or conical hull of a finite set is closed.

Proof. We will prove the result for conical hulls; the result for convex hulls follows using (2.1.5). We induct on the dimension of V. Let $S \subset V = \mathbb{R}^n$ be finite. By Carathéodory's Theorem, $\operatorname{cone} S$ is a union of cones of the form $\operatorname{cone} T$, where T ranges over all subsets of S with size at most n. Let T be a subset of S with size at most n. If T is contained in a proper linear subspace of V, then $\operatorname{cone} T$ is closed by the inductive hypothesis. Otherwise, T is a set of n linearly independent vectors. Then $\operatorname{cone} T$ is a simplicial cone, and hence is a polyhedron and therefore closed. Since S is finite, there are finitely many such T, and so $\operatorname{cone} S$ is closed. $\square$

We will later prove that the convex or conical hull of a finite set is in fact a polyhedron (Theorem 2.5.3).

Interior and boundary

Recall that the *interior* of a set $S \subset V$ is the set of all $x \in S$ such that there exists an open set U of V with $x \in U \subset S$. We denote the interior of S by $\operatorname{int} S$. The *boundary* of S, denoted $\operatorname{bd} S$, is defined as $\operatorname{cl} S \smallsetminus \operatorname{int} S$, where $\operatorname{cl} S$ is the closure of S in V.

The definitions of interior and boundary depend on the ambient space V. If S is not full-dimensional, then we always have $\operatorname{int} S = \varnothing$ and $\operatorname{bd} S = \operatorname{cl} S$. In many cases it is more useful to use a concept of interior and boundary which is intrinsic to S. We define the *relative interior* and *relative boundary* of S to be its interior and boundary with respect to the subspace topology on $\operatorname{aff} S$. We denote them $\operatorname{rint} S$ and $\operatorname{rbd} S$, respectively. The relative interior of any nonempty convex set is nonempty (Exercise 2.5).

Example 2.1.8. Let $\Delta \subset \mathbb{R}^{n+1}$ be the simplex from Example 1.2.8. We have

$$\begin{aligned} \operatorname{int}\Delta &= \varnothing & \qquad \operatorname{rint}\Delta &= \Delta \cap \{e^i(x) > 0,\ i = 1, \ldots, n+1\} \\ \operatorname{bd}\Delta &= \Delta & \qquad \operatorname{rbd}\Delta &= \Delta \cap \bigcup_{i=1}^{n+1} \{e^i(x) = 0\} \end{aligned}$$

The relative interior of Δ can also be characterized as the set of all *positive convex combinations* of $e_1, \ldots, e_{n+1}$; that is, all convex combinations $\sum_{i=1}^{n+1} \lambda_i e_i$ where $\lambda_i > 0$ for all i.

Lineality space and recession cone

For a set $S \subset V$, we define the *lineality space* and *recession cone* of S to be the following sets, respectively:

$$\operatorname{lins} S := \begin{cases} \{v \in V : x + \lambda v \in S \text{ for all } x \in S,\ \lambda \in \mathbb{R}\} & \text{if } S \neq \varnothing \\ \{0\} & \text{if } S = \varnothing \end{cases}$$

$$\operatorname{recc} S := \begin{cases} \{v \in V : x + \lambda v \in S \text{ for all } x \in S,\ \lambda \geq 0\} & \text{if } S \neq \varnothing \\ \{0\} & \text{if } S = \varnothing \end{cases}.$$

If K is convex, then $\operatorname{lins} K$ is a vector subspace and $\operatorname{recc} K$ is a convex cone (Exercise 2.7). If K is bounded, then $\operatorname{lins} K = \operatorname{recc} K = \{0\}$. Lineality spaces and recession cones describe the behavior of unbounded convex sets at infinity. Exercises 2.8–2.10 cover some basic properties of these sets. We highlight the following in particular (Exercise 2.10).

Proposition 2.1.9. Let $K = \{y_i(x) \le b_i,\, i \in I\}$ where I is any set and $y_i \in V^*$, $b_i \in \mathbb{R}$ for all i. Assume K is nonempty. Then

$$\begin{aligned} \operatorname{lins} K &= \{y_i(x) = 0,\, i \in I\} \\ \operatorname{recc} K &= \{y_i(x) \le 0,\, i \in I\}. \end{aligned}$$

Let $K \subset V$ be a convex set and let $L = \operatorname{lins} K$. Let $q : V \to V/L$ be the quotient map with kernel L. Then $q(K) = K/L$ is a convex set in V/L with lineality space $\{0\}$. This is a way to reduce questions about general convex sets to convex sets with trivial lineality space. Alternatively, choose any linear subspace W of V such that $V = W \oplus L$. Then we can write $K = K' \oplus L$, where $K' := K \cap W$ is a convex set with $\operatorname{lins} K' = \{0\}$.

2.2 Support and separation

In this section we prove the *supporting hyperplane theorem* and *separating hyperplane theorem*, two fundamental theorems in geometry. Our proof is an inductive argument. We also give a standard metric argument in Exercise 2.12.

The following is the key lemma.

Theorem 2.2.1. Let $C \subset V$ be a convex cone with $C \ne V$. Then there exists $y \in V^* \smallsetminus \{0\}$ such that $y(x) \le 0$ for all $x \in C$.

Given a set $S \subset V$, a point $x \in S$, and an affine hyperplane H of V, we say that H *supports* S at x if $x \in H$ and S lies in one of the closed half-spaces bounded by H. Theorem 2.2.1 says that any convex cone which is not the entire vector space is supported by a hyperplane at 0.

Proof of Theorem 2.2.1. We induct on $\dim V$. We leave the case $\dim V \le 2$ as an exercise (Exercise 2.13). Assume $\dim V \ge 3$. Let H be a linear hyperplane of V which contains a vector not in C. Then $C' := C \cap H$ is a convex cone in H with $C' \ne H$. By the inductive hypothesis, there is a hyperplane H' of H that supports C' at 0. Consider the quotient map $q : V \to V/H'$. Then $q(C)$ is a convex cone in V/H', and $\dim(V/H') = 2$. Since $C \cap (x + H') = \varnothing$ for any x in H on the opposite side of H' as C', we have $q(C) \neq V/H'$. Thus, by the inductive hypothesis, there is a hyperplane H'' of V/H' that supports $q(C)$ at 0. Then $q^{-1}(H'')$ is a hyperplane of V that supports C at 0, as desired. $\square$

Recall that for a set $S \subset V$, the set $S^\perp \subset V^*$ is the set of all linear functionals which are constant on S. We have the following slight generalization of Theorem 2.2.1.

Corollary 2.2.2. Let $C \subset V$ be a convex cone which is not a linear subspace. Then there exists $y \in V^* \setminus C^\perp$ such that $y(x) \leq 0$ for all $x \in C$.

Proof. This follows from applying Theorem 2.2.1 to C when viewed as a cone in the vector space $\operatorname{span} C$. $\square$

We can now easily prove the supporting and separating hyperplane theorems.

Theorem 2.2.3 (Supporting hyperplane theorem)**.** Let K be a convex subset of V and let $x_0 \in \operatorname{rbd} K$. Then there exists $y \in V^* \setminus K^\perp$ such that $y(x) \leq y(x_0)$ for all $x \in K$.

Proof. For any $x \in K$, define the *direction cone* of K at x to be the set

$$D(K, x) := \operatorname{cone}(K - x).$$

(This is also called the *cone of feasible directions*.) Since $x_0 \in \operatorname{rbd} K$, we have that $D(K, x_0)$ is not a linear subspace. Hence, by Corollary 2.2.2, there exists $y \in V^* \setminus D(K, x_0)^\perp = V^* \setminus K^\perp$ such that $y(v) \leq 0$ for all $v \in D(K, x_0)$. Then $y(x - x_0) \leq 0$ for all $x \in K$. $\square$

Theorem 2.2.4 (Separating hyperplane theorem)**.** Let K be a nonempty closed convex subset of V and let $x_0 \in V \setminus K$. Then there exists $y \in V^*$ which achieves a maximum on K and $\max_K y < y(x_0)$.

Proof. If $x_0 \notin \operatorname{aff} K$, then we can find a hyperplane containing K and not x_0, and the result follows. Thus, assume $x_0 \in \operatorname{aff} K$. Let $u \in \operatorname{rint} K$. The open line segment (u, x_0) contains a point $v \in \operatorname{rbd} K$, which is in K since K is closed. Applying the supporting hyperplane theorem to K at v gives the desired y. $\square$

Corollary 2.2.5. If K is a nonempty closed convex set in V, then

$$K = \{x \in V : y(x) \leq \max_K y \text{ for all } y \text{ achieving a maximum on } K\}.$$

In particular, this shows that every closed convex set has a constraint presentation.

When applied to cones, the separating hyperplane theorem has a useful corollary known as Farkas' lemma.

Corollary 2.2.6 (Farkas' lemma)**.** Let C be a convex cone in V and let $x_0 \in V \setminus \operatorname{cl} C$. Then there exists $y \in V^* \setminus \{0\}$ such that $y(x_0) > 0$ and $y(x) \leq 0$ for all $x \in C$.

Proof. Applying the separating hyperplane theorem to $\operatorname{cl} C$ and x_0, there exists $y \in V^* \setminus \{0\}$ such that $\sup_C y < y(x_0)$ for all $x \in C$. The result follows by Lemma 1.2.2. □

2.3 Polar cones

Let $S \subset V$. The *polar cone* of S is the set $S^\circ \subset V^*$ defined by

$$S^\circ = \{y \in V^* : y(x) \leq 0 \text{ for all } x \in S\}.$$

For a fixed $x \in V$, the set $\{y \in V^* : y(x) \leq 0\}$ is a closed half-space in V^*. The set S° is the intersection of all such half-spaces over all $x \in S$, and hence is a closed convex cone in V^*. Note the shift in perspective: Whereas before we fixed $y \in V^*$ and viewed $y(x) \leq 0$ as a linear constraint on x, we now fix $x \in V$ and view $y(x) \leq 0$ as a linear constraint on y.

Theorem 2.3.1. For all $S \subset V$, we have $S^{\circ\circ} = \operatorname{cl} \operatorname{cone} S$. In particular, $S^{\circ\circ} = S$ if S is a closed convex cone.

Proof. Suppose $x_0 \in \operatorname{cone} S$. Let $y \in S^\circ$. Since $y(x) \leq 0$ for all $x \in S$, by linearity we have $y(x_0) \leq 0$. This holds for all $y \in S^\circ$, so $x_0 \in S^{\circ\circ}$. Hence $\operatorname{cone} S \subset S^{\circ\circ}$. Since $S^{\circ\circ}$ is closed as noted above, we have $\operatorname{cl} \operatorname{cone} S \subset S^{\circ\circ}$.

Now suppose $x_0 \in V \setminus \operatorname{cl} \operatorname{cone} S$. By Farkas' lemma, there exists $y \in V^*$ such that $y(x_0) > 0$ and $y(x) \leq 0$ for all $x \in S$. The latter inequality implies $y \in S^\circ$. Then since $y(x_0) > 0$, we have $x_0 \notin S^{\circ\circ}$, as desired. □

As noted in the next proposition, polarity connects two ways to present a convex cone: As the conical hull of a set of vectors, or as the set of solutions to a system of linear inequalities.

Proposition 2.3.2. The following are true.

(a) Let $C = \operatorname{cone} S$ where $S \subset V$. Then

$$C^\circ = S^\circ = \{y \in V^* : y(x) \le 0 \text{ for all } x \in S\}.$$

(b) Let

$$C = T^\circ = \{x \in V : y(x) \le 0 \text{ for all } y \in T\}$$

where $T \subset V^*$. Then $C^\circ = \operatorname{cl}\operatorname{cone} T$.

Proof. Proof of (a). Clearly $C^\circ \subset S^\circ$. If $y \in S^\circ$, then $y(x) \le 0$ for all $x \in S$, so by linearity $y(x) \le 0$ for all $x \in \operatorname{cone} S = C$. Hence $S^\circ \subset C^\circ$.

Proof of (b). This follows from Theorem 2.3.1. □

Proposition 2.3.3. Let $C \subset V$. Then $\operatorname{span} C$ and $\operatorname{lins}(C^\circ)$ are orthogonal complements. If C is a closed convex cone, then $\operatorname{lins} C$ and $\operatorname{span}(C^\circ)$ are orthogonal complements.

Proof. Exercise 2.15. □

Balanced support

Closed convex cones satisfy a stronger support property than Theorem 2.2.1. This property will be important in the next chapter.

Given a convex cone $C \subset V$, we say that C has *balanced support* if there exists $y \in V^*$ such that $\operatorname{argmax}_C y = \operatorname{lins} C$. By Lemma 1.2.2, the latter equality holds if and only if $y(x) \le 0$ for all $x \in C$ (that is, $y \in C^\circ$), and

$$\{x \in C : y(x) = 0\} = \operatorname{lins} C. \tag{2.3.4}$$

For example, $C = V$ has balanced support with $y = 0$. If $C \ne V$, then C has balanced support if and only if it is supported by a hyperplane H at 0 and $C \cap H = \operatorname{lins} C$.

Theorem 2.3.5. Let $C \subset V$ be a closed convex cone and $y \in V^*$. Then $\operatorname{argmax}_C y = \operatorname{lins} C$ if and only if $y \in \operatorname{rint} C^\circ$. In particular, every closed convex cone has balanced support.

Proof. For the "if" direction, suppose $y \in \operatorname{rint} C^\circ$. Since $y \in C^\circ$, we only need to show (2.3.4). By Proposition 2.3.3, $\operatorname{lins} C$ and $\operatorname{span} C^\circ$ are orthogonal complements. In particular $\operatorname{lins} C$ and y are orthogonal, so $\operatorname{lins} C \subset \{x \in C : y(x) = 0\}$. For the other inclusion, suppose $x \in C$

and $y(x) = 0$. Since $x \in C = C^{\circ\circ}$, we have $y'(x) \le 0$ for all $y' \in C^\circ$. So by Exercise 2.6, we have $y'(x) = 0$ for all $y' \in C^\circ$. Again since $\operatorname{lins} C$ and $\operatorname{span} C^\circ$ are orthogonal complements, this means $x \in \operatorname{lins} C$ as desired.

For the "only if" direction, suppose $y \notin \operatorname{rint} C^\circ$. If $y \notin C^\circ$ then y is unbounded on C, so we are done. Thus assume $y \in \operatorname{rbd} C^\circ$. By the supporting hyperplane theorem applied to C° at y, there exists $x \in V \setminus (C^\circ)^\perp$ such that $y(x) = 0$ and $y'(x) \le 0$ for all $y' \in C^\circ$. The latter statement implies $x \in C^{\circ\circ} = C$. In addition $(C^\circ)^\perp = \operatorname{lins} C$, so $x \in C \setminus \operatorname{lins} C$. Hence there is some $x \in C \setminus \operatorname{lins} C$ such that $y(x) = 0$, so (2.3.4) does not hold. □

2.4 Extreme rays and points

Faces

Given a convex set K, a *face* of K is a nonempty convex set $F \subset K$ such that if $x \in F$ and x is a positive convex combination of $x_1, \ldots, x_m \in K$, then $x_1, \ldots, x_m \in F$. For cones, we can replace "convex set" with "convex cone" and "convex combination" with "conical combination" for an equivalent definition.

Example 2.4.1. If K is a convex set and y is a linear functional which achieves a maximum on K, then $\operatorname{argmax}_K y$ is a face of K. A face which arises in this way is called a *support set* of K. If K is nonempty, we have $K = \operatorname{argmax}_K 0$, so K is a support set of K.

It follows from the definition that if F is a face of K and $F' \subset F$, then F' is a face of F if and only if F' is a face of K.

In this text, a *ray* will mean a cone of dimension 1 (so the endpoint is at 0). Any nonzero vector in a ray is a *generator* for the ray. We define *extreme points* and *extreme rays* to be faces which are points and rays, respectively. Extreme points of polyhedra are called *vertices*.

Extremal representations

We prove the following major theorem: For a closed convex cone with trivial lineality space, the extreme rays provide the unique minimal way to write the cone as a conical hull.

Theorem 2.4.2. Let $C \subset V$ be a closed convex cone with $\operatorname{lins} C = \{0\}$. Let $S \subset C$. Then $C = \operatorname{cone} S$ if and only if S contains a generator for every extreme ray of C.

Proof. Suppose $C = \operatorname{cone} S$. If S does not contain a generator for some extreme ray R, then the definition of a face implies $R \not\subset \operatorname{cone} S$, a contradiction. Thus S must contain a generator for every extreme ray.

For the other direction, it suffices to show that C equals the conical hull of its extreme rays. We induct on the dimension of V. If $\dim V \leq 1$, then C is either $\{0\}$ or a single ray, and the statement holds. Assume $\dim V \geq 2$. Let $x \in C$. Since C is closed and $\operatorname{lins} C = \{0\}$, by Theorem 2.3.5, C is supported by a hyperplane H with $C \cap H = \{0\}$. Hence, there exists $v \in V$ such that v, $-v \notin C$. Since C is closed, we have that $x + \lambda v$ and $x - \lambda v$ are not in C for large enough $\lambda > 0$ (Exercise 2.9). Thus there exist λ_1, $\lambda_2 \geq 0$ such that $x_1 := x + \lambda_1 v$ and $x_2 := x - \lambda_2 v$ are in the boundary of C. Let H be a supporting hyperplane of C at x_1. Then $C \cap H$ is a closed convex cone in H with lineality space $\{0\}$. Applying the inductive hypothesis, x_1 is in the conical hull of the extreme rays of $C \cap H$. Moreover, $C \cap H$ is a face of C by Example 2.4.1, so any extreme ray of $C \cap H$ is an extreme ray of C. Hence x_1 is in the conical hull of the extreme rays of C. The same argument holds for x_2. Since x is a conical combination of x_1 and x_2, it follows that x is in the conical hull of the extreme rays of C. $\square$

Corollary 2.4.3. Let $K \subset V$ be a bounded closed convex set. Let $S \subset K$. Then $K = \operatorname{conv} S$ if and only if S contains every extreme point of K.

Proof. Let $C = \operatorname{cone}(K \times \{1\}) \subset V \times \{1\}$. The extreme rays of C are precisely $\operatorname{cone}(\{(x, 1)\})$ over all extreme points x of K (Exercise 2.19). Moreover, since K is bounded and closed, C is closed and has lineality space $\{0\}$. The result then follows from Theorem 2.4.2. $\square$

Finite representations of polyhedra

When restricted to polyhedra, the previous theorem implies that polyhedral cones/polytopes can be characterized as conical/convex hulls of finite sets.

Proposition 2.4.4. The following are true.
(a) A polyhedral cone has finitely many extreme rays.
(b) A polyhedron has finitely many vertices.

Proof. Proof of (a). Let C be a polyhedral cone. If $\operatorname{lins} C \neq \{0\}$ then C has no extreme rays. (See Exercise 2.18.) So assume $\operatorname{lins} C = \{0\}$. We induct on the dimension of V. It is obvious for $\dim V \leq 1$. Assume $\dim V \geq 2$. Write $C = \{y_i(x) \leq 0,\ i \in I\}$ for some finite set I. We may assume $y_i \neq 0$ for all i. Let $H_i = \{y_i(x) = 0\}$. If an element of C is not in any of the H_i, then it is in the set $\{y_i(x) < 0,\ i \in I\}$, which is the interior of C. Since $\dim V \geq 2$, a ray containing an interior element of C cannot be an extreme ray. Thus every extreme ray of C is contained in some H_i. An extreme ray of C contained in H_i is also an extreme ray of $C \cap H_i$. Each $C \cap H_i$ is a polyhedral cone in H_i with trivial lineality space, so by the inductive hypothesis it has finitely many extreme rays. There are finitely many H_i, so C has finitely many extreme rays.

Proof of (b). This follows from a similar argument. □

We now state our main result.

Theorem 2.4.5. The following are true.

(a) A subset of V is a polyhedral cone if and only if it is the conical hull of a finite set of vectors.

(b) A subset of V is a polytope if and only if it is the convex hull of a finite set of points.

Proof. Proof of (a). For the "only if" direction, let C be a polyhedral cone. Write $C = C' \oplus L$ where $L = \operatorname{lins} C$ and C' is a polyhedral cone with $\operatorname{lins} C' = \{0\}$. Since L is a vector space, it is the conical hull of a finite set. By Theorem 2.4.2 and Proposition 2.4.4, C' is the conical hull of a finite set. Hence C is the conical hull of a finite set.

For the "if" direction, let C be the conical hull of a finite set. By Proposition 2.3.2(a), C° is a polyhedral cone in V^*. Then by what we just proved, C° is the conical hull of a finite set. By Proposition 2.3.2(a) again, $C^{\circ\circ}$ is a polyhedral cone. Since C is closed by Proposition 2.1.7, we have $C = C^{\circ\circ}$, proving the result.

Proof of (b). The "only if" direction is Corollary 2.4.3 and Proposition 2.4.4. The "if" direction follows by part (a) and equation (2.1.5). □

Corollary 2.4.6. The polar cone of a polyhedral cone is polyhedral.

Proof. This follows from Theorem 2.4.5 and Proposition 2.3.2. □

2.5 Minkowski's theorem

Given two sets S, $T \subset V$, the *Minkowski sum* of S and T is the set

$$S + T := \{s + t : s \in S, t \in T\}.$$

If either S or T are empty then $S + T = \varnothing$. It is easy to show that if S and T are convex then $S + T$ is convex.

Example 2.5.1. The cube from Example 1.2.6 is the Minkowski sum of the line segments $[-e_i, e_i]$ over all $i = 1, \ldots, n$.

Theorem 2.4.2 and Corollary 2.4.3 concerned writing closed convex cones and bounded sets in terms of their extreme rays and points. For general closed convex sets, we have the following.

Theorem 2.5.2 (Minkowski)**.** Let K be a closed convex set with $\operatorname{lins} K = \{0\}$. Let S be the set of extreme points of K and T be the set of extreme rays of $\operatorname{recc} K$. Then $K = \operatorname{conv} S + \operatorname{cone} T$.

Similar to Corollary 2.4.3, we will prove this theorem by looking at $\operatorname{cone}(K \times \{1\})$. However, if K is unbounded, then $\operatorname{cone}(K \times \{1\})$ will not be closed. We can fix this as follows. Given a convex set $K \subset V$, define its *homogenization* $\operatorname{homog} K$ to be the set in $V \times \mathbb{R}$ given by

$$\operatorname{homog} K := \operatorname{cone}(K \times \{1\}) \cup (\operatorname{recc} K \times \{0\}).$$

Equivalently, $\operatorname{homog} K$ is $\operatorname{cone}(K \times \{1\})$ along with all limit points of $\operatorname{cone}(K \times \{1\})$ in $V \times \{0\}$ (Exercise 2.20). The set $\operatorname{homog} K$ is a convex cone which is closed if K is closed, and polyhedral if K is polyhedral. This and other properties can be found in Exercises 2.21–2.24. We use these to sketch a proof of Theorem 2.5.2.

Proof of Theorem 2.5.2. By Exercise 2.21, $\operatorname{homog} K$ is a closed convex cone with lineality space $\{0\}$. Thus it is the conical hull of its extreme rays. By Exercise 2.23, the set of extreme rays of $\operatorname{homog} K$ is $S \times \{0\} \cup T \times \{1\}$. It follows that $K = \operatorname{conv} S + \operatorname{cone} T$, by Exercise 2.22. $\square$

Minkowski's theorem for polyhedra

We can give an extension of Theorem 2.4.5 to general polyhedra.

Theorem 2.5.3. A subset P of V is a polyhedron if and only if $P = \operatorname{conv} S + \operatorname{cone} T$ for some finite sets S, $T \subset V$. If this is the case and P is nonempty, then $\operatorname{recc} P = \operatorname{cone} T$.

Proof. For the "only if" direction, let P be a polyhedron, and write $P = P' \oplus L$ where $L = \operatorname{lins} P$ and P' is a polyhedron with $\operatorname{lins} P' = \{0\}$. Since L is the conical hull of a finite set, it suffices to prove the result for P', so we may assume $P = P'$. We have that $\operatorname{recc} P$ is polyhedral by Proposition 2.1.9. Thus the result follows from Theorem 2.5.2 and Proposition 2.4.4.

For the "if" direction, suppose $P = \operatorname{conv} S + \operatorname{cone} T$ with S, $T \subset V$ finite. By Exercise 2.22, $\operatorname{homog} P$ is the conical hull of a finite set. Hence it is polyhedral by Theorem 2.4.5, so P is polyhedral as well.

The second part of the theorem follows from Exercise 2.25. □

We conclude by using Theorem 2.5.3 to prove the following statements.

Theorem 2.5.4. If $f : V \to W$ is an affine map and $P \subset V$ is a polyhedron, then $f(P)$ is a polyhedron.

Proof. An affine map $f : V \to W$ can be written as $f(x) = f_0(x) + b$, where $f_0 : V \to W$ is a linear map and $b \in \mathbb{R}$. It is straightforward to check that

$$f(\operatorname{conv} S + \operatorname{cone} T) = \operatorname{conv} f(S) + \operatorname{cone} f_0(T).$$

□

Theorem 2.5.5. The Minkowski sum of two polyhedra is a polyhedron.

Proof. Let $P_1 = \operatorname{conv} S_1 + \operatorname{cone} T_1$ and $P_2 = \operatorname{conv} S_2 + \operatorname{cone} T_2$ be polyhedra, where S_i, T_i are finite for $i = 1, 2$. Then by Exercise 2.26,

$$P_1 + P_2 = \operatorname{conv}(S_1 + S_2) + \operatorname{cone}(T_1 \cup T_2)$$

so $P_1 + P_2$ is a polyhedron. □

Theorem 2.5.6. If P, Q are polyhedra, then $\operatorname{conv}(P \cup Q)$ is a polyhedron.

Proof. Exercise 2.27. □

Theorem 2.5.7. Let P be a nonempty polyhedron and $y \in V^*$ a linear functional.
(a) If $y \in (\operatorname{recc} P)^\circ$, then y achieves a maximum on P.
(b) If $y \notin (\operatorname{recc} P)^\circ$, then y is unbounded from above on P.

Proof. By Theorem 2.5.3, we have $P = \operatorname{conv} S + \operatorname{recc} P$ for some nonempty finite set S. If $y \in (\operatorname{recc} P)^\circ$, then y achieves a maximum of 0 on $\operatorname{recc} P$, so y achieves a maximum of $\max_S y$ on P. If $y \notin (\operatorname{recc} P)^\circ$, then $y(v) > 0$ for some $v \in \operatorname{recc} P$, and thus $y(x + \lambda v) \to \infty$ for $x \in P$ and $\lambda \to \infty$. $\square$

Exercises

2.1. The convex hull of a compact set is compact.
2.2. Proposition 2.1.7 is not true if we replace "finite" with "compact".
2.3. If S is a compact and $0 \notin \operatorname{conv} S$, then $\operatorname{cone} S$ is closed.
2.4. The closure, interior, and relative interior of a convex set are convex.
2.5. The relative interior of any nonempty convex set is nonempty.
2.6. Let K be a convex set and $x_0 \in \operatorname{rint} K$. Suppose $y \in V^*$ such that $y(x) \leq y(x_0)$ for all $x \in K$. Then $y(x) = y(x_0)$ for all $x \in K$.
2.7. Let K be a convex subset of V. Then $\operatorname{lins} K$ is a vector subspace of V, and $\operatorname{recc} K$ is a convex cone.
2.8. (a) Let K be a convex set. Then

$$\operatorname{lins} K = \operatorname{lins} \operatorname{recc} K = (\operatorname{recc} K) \cap (-\operatorname{recc} K).$$

(b) If K is a convex cone, then $\operatorname{recc} K = K$.
2.9. Let K be a closed convex subset of V. For a nonzero vector $v \in V$, the following two statements are equivalent:
(i) $v \in \operatorname{lins} K$.
(ii) For some $x_0 \in K$, we have $x_0 + \lambda v \in K$ for all $\lambda \in \mathbb{R}$.
In addition, the following two statements are equivalent:
(iii) $v \in \operatorname{recc} K$.
(iv) For some $x_0 \in K$, we have $x_0 + \lambda v \in K$ for all $\lambda \geq 0$.
2.10. Let $K = \{y_i(x) \leq b_i,\ i \in I\}$ where I is any set and $y_i \in V^*$, $b_i \in \mathbb{R}$ for all i. Assume K is nonempty. Then

$$\begin{aligned} \operatorname{lins} K &= \{y_i(x) = 0,\ i \in I\} \\ \operatorname{recc} K &= \{y_i(x) \leq 0,\ i \in I\}. \end{aligned}$$

2.11. The Minkowski sum of two closed convex sets is not necessarily closed. *Hint:* Consider the sets $\{x \in \mathbb{R}^2 : e^1(x)e^2(x) \geq 1\}$ and $\{x : \mathbb{R}^2 : e^1(x) \geq 0\}$.
2.12. We give a metric proof of Theorem 2.2.4. Let K be a nonempty closed convex subset of V and let $u \in V \setminus \operatorname{cl} K$. Fix an inner product $\langle \cdot, \cdot \rangle$ and norm $||x|| = \langle x, x \rangle$.
(a) There exists $v \in K$ such that $||u-v||$ is the minimum distance between u and any point in K.
(b) Let $y = \langle u-v, \cdot \rangle$ and $b = y(v)$. Then $y(u) > b$ and $y(x) \leq b$ for $x \in K$.
2.13. Let $\dim V \leq 2$, and let $C \subset V$ be a convex cone with $C \neq V$. Then there exists $y \in V^* \setminus \{0\}$ such that $y(x) \leq 0$ for all $x \in C$.
2.14. Let K and L be two disjoint nonempty closed convex sets in V such that $\operatorname{recc} K \cap \operatorname{recc} L = \{0\}$. Then there exist $y \in V^* \setminus \{0\}$ and $b \in \mathbb{R}$ such that $y(x) > b$ for all $x \in K$ and $y(x) < b$ for all $x \in L$.
Hint: Write $K = \{y_i(x) \leq b_i : i \in I\}$ and $L = \{z_j(x) \leq c_j : j \in J\}$. Apply Theorem 2.2.1 to the conical hull of $\{(y_i, -b_i)\} \cup \{(z_j, -b_j)\}$ in $V^* \times \mathbb{R}$.
2.15. Let $C \subset V$. Then $\operatorname{span} C$ and $\operatorname{lins}(C^\circ)$ are orthogonal complements. If C is a closed convex cone, then $\operatorname{lins} C$ and $\operatorname{span}(C^\circ)$ are orthogonal complements.
2.16. Let $C \subset V$.
(a) If $C = (\operatorname{cone} S) + L$ where $S \subset V$ and L is a vector subspace, then

$$C^\circ = \{y \in V^* : y(x) \leq 0 \text{ for all } x \in S\} \cap L^\perp$$

(b) Suppose

$$C = \{x \in V : y(x) \leq 0 \text{ for all } y \in T\} \cap L$$

where $T \subset V^*$, $\operatorname{cone} T$ is closed, and L is a vector subspace. Then $C^\circ = (\operatorname{cone} T) + L^\perp$.
2.17. Let K be a convex set. A nonempty convex set $F \subset K$ is a face of K if and only for every $x \in K$, if $x = \frac{1}{2}(x_1 + x_2)$ where $x_1, x_2 \in K$, then $x_1, x_2 \in F$.
2.18. Let K be a convex set. Every face of K contains a translation of $\operatorname{lins} K$.
2.19. Let $K \subset V$ be a convex set. The extreme rays of $\operatorname{cone}(K \times \{1\}) \subset V \times \mathbb{R}$ are exactly the cones $\operatorname{cone}(\{(x, 1)\}$ over all extreme points x of K.
2.20. Let $K \subset V$ be a convex set. Then $\operatorname{homog} K$ is the union of $\operatorname{cone}(K \times \{1\})$ with the set of limits points of $\operatorname{cone}(K \times \{1\})$ in $V \times \{0\}$.
2.21. Let K be a convex set. The following are true.

(a) $\operatorname{homog} K$ is a convex cone. If K is closed, then so is $\operatorname{homog} K$.
(b) $\operatorname{lins}(\operatorname{homog} K) = \operatorname{lins} K \times \{0\}$.

2.22. Let K be a convex set in V. If $S, T \subset V$ such that

$$\operatorname{homog} K = \operatorname{cone}((S \times \{1\}) \cup (T \times \{0\})),$$

then $K = \operatorname{conv} S + \operatorname{cone} T$. The converse holds if S is bounded.

2.23. The extreme rays of $\operatorname{homog} K$ are exactly the rays $\operatorname{cone}\{(x,1)\}$ over all extreme points x of K, along with the extreme rays of $\operatorname{recc} K \times \{0\}$.

2.24. Let $K = \{y_i(x) \le b_i,\ i \in I\}$ where I is any set. Assume K is nonempty. Then

$$\operatorname{homog} K = \{(x,t) \in V \times \mathbb{R} : t \ge 0,\ y_i(x) - b_i t \le 0 \text{ for all } i \in I\}.$$

2.25. Let B be a nonempty bounded convex set and C a closed convex cone. Then

$$\begin{aligned} \operatorname{recc}(B + C) &= C \\ \operatorname{homog}(B + C) &= \operatorname{cone}((B \times \{1\}) \cup (C \times \{0\})). \end{aligned}$$

2.26. For any two sets $S_1, S_2 \subset V$,

$$\begin{aligned} \operatorname{cone}(S_1 \cup S_2) &= \operatorname{cone}(S_1 \cup S_2) \\ \operatorname{conv}(S_1) + \operatorname{conv}(S_2) &= \operatorname{conv}(S_1 + S_2). \end{aligned}$$

2.27. If P, Q are polyhedra, then $\operatorname{conv}(P \cup Q)$ is a polyhedron.

Chapter 3

Faces

3.1 Faces of convex sets and polyhedra

Faces of convex sets

Let $K \subset V$ be a convex set. Recall that a *face* of K is a nonempty convex set $F \subset K$ such that if $x \in F$ and x is a positive convex combination of $x_1, \ldots, x_m \in K$, then $x_1, \ldots, x_m \in F$. Equivalently, a nonempty convex set $F \subset K$ is a face of K if and only for every $x \in F$, if $x = \frac{1}{2}(x_1 + x_2)$ for some $x_1, x_2 \in K$, then $x_1, x_2 \in F$ (Exercise 2.17). If K is nonempty, then K is a face of itself. A *proper face* of K is a face which is not K. If $K = \varnothing$, then K has no faces.

We begin by proving a general characterization of faces. Recall that for a point $x \in K$, the *direction cone* of K at x is

$$D(K, x) = \operatorname{cone}(K - x).$$

In other words, $v \in D(K, x)$ if and only if $x + \lambda v \in K$ for some $\lambda > 0$. By Exercise 2.8, we have

$$\operatorname{lins} D(K, x) = D(K, x) \cap (-D(K, x)).$$

In other words, $v \in \operatorname{lins} D(K, x)$ if and only if $x + \lambda v \in K$ for some $\lambda > 0$ and some $\lambda < 0$.

Proposition 3.1.1. Let K be a convex set, and let $x \in K$. The following are true.

(a) There is exactly one face F of K such that $x \in \operatorname{rint} F$, and for any face G of K containing x, we have $G \supset F$.
(b) The face F from part (a) is given by

$$F = K \cap (x + \operatorname{lins} D(K, x)).$$

In addition, $\operatorname{aff} F = x + \operatorname{lins} D(K, x)$.

Proof. Suppose F, G are faces such that $x \in \operatorname{rint} F$ and $x \in G$. Let $x_1 \in F$. Since $x \in \operatorname{rint} F$ and F is convex, there is $x_2 \in F$ such that x is in the relative interior of the segment $[x_1, x_2]$. Since G is face, we must therefore have $x_1 \in G$. Hence $G \supset F$, proving our claim that any face containing x must contain F. If furthermore $x \in \operatorname{rint} G$, then we similarly have $F \supset G$, so $F = G$. This proves that x is in the relative interior of at most one face.

Now let $L = \operatorname{lins} D(K, x)$ and let $F = K \cap (x + L)$. To complete the proof of the theorem, it suffices to show that F is a face of K, that $x \in \operatorname{rint} F$, and $\operatorname{aff} F = x + L$. We show the latter two statements first. By definition F is contained in the affine space $x + L$. Also, by definition of L, we have that K contains a neighborhood of x in the relative topology of $x + L$, and hence F does as well. Hence $\operatorname{aff} F = x + L$, and $x \in \operatorname{rint} F$.

We now prove F is a face. Since F is the intersection of two convex sets, it is convex. Let $x' \in F$ and suppose $x' = \frac{1}{2}(x_1 + x_2)$ for some x_1, $x_2 \in K$. Since $x' \in F$ and $x \in \operatorname{rint} F$, there is $x_3 \in F$ such that x is in the relative interior of the segment $[x', x_3]$. Then x is in the relative interior of $T := \operatorname{conv}(x_1, x_2, x_3)$. Since $T \subset K$ by convexity, we have $T - x \subset L$ by definition of L. Thus $T \subset x + L$, so $T \subset F$. In particular x_1, $x_2 \in F$, so F is a face of K. $\square$

Corollary 3.1.2. Every convex set is the disjoint union of the relative interiors of its faces.

Corollary 3.1.3. Every face of a closed convex set is closed, and every face of a polyhedron is polyhedral.

If x, $x' \in \operatorname{rint} F$ for some face F of K, then $D(K, x) = D(K, x')$ (Exercise 3.4). We can therefore define $D(K, F)$ to be $D(K, x)$ for any $x \in \operatorname{rint} F$. By Proposition 3.1.1, $\operatorname{lins} D(K, F)$ is the linear subspace parallel to $\operatorname{aff} F$.

Support sets

Let K be a convex set. Recall from Example 2.4.1 that a *support set* of K is a set of the form $\operatorname{argmax}_K y$ where $y \in V^*$ is a linear functional which achieves a maximum on K. Support sets are also called *exposed faces*. All support sets are faces, but not every face is a support set (Exercise 3.8). The precise condition for a face to be a support set is as follows.

Proposition 3.1.4. Let K be a convex set and F a face of K. Then F is a support set of K if and only if $D(K, F)$ has balanced support.

This is part (b) of the following proposition (Exercise 3.9).

Proposition 3.1.5. Let K be a convex set and F a face of K. Let $y \in V^*$. The following are true.
(a) $F \subset \operatorname{argmax}_K y$ if and only if $y \in D(K, F)^\circ$.
(b) $F = \operatorname{argmax}_K y$ if and only if $\operatorname{argmax}_{D(K,F)} y = \operatorname{lins} D(K, F)$.

Faces of polyhedra

Let P be a polyhedron. By Proposition 3.1.1, every face of P is the intersection of P with an affine space, and is therefore a polyhedron. We start by showing the following.

Proposition 3.1.6. Let $P = \{y_i(x) \le b_i,\ i \in I\}$ be a polyhedron, where I is finite. Let F be a face of P, and let $J \subset I$ be the set of all j such that $y_j(x) = b_j$ for all $x \in F$. Then

$$D(P, F) = \{y_j(x) \le 0,\ j \in J\}.$$

Proof. Let $x_0 \in \operatorname{rint} F$. By Exercise 2.6, we have $j \in J$ if and only if $y_j(x_0) = b_j$. It is easy to check that $D(P, x_0) \subset \{y_j(x) \le 0,\ j \in J\}$. For the reverse inclusion, let $v \in \{y_j(x) \le 0,\ j \in J\}$. By the definition of J, we have $y_i(x_0) < b_i$ for all $i \in I \setminus J$. Since $I \setminus J$ is finite, we can choose $\lambda > 0$ small enough so that $y_i(x_0 + \lambda v) < b_i$ for all $i \in I \setminus J$. Moreover, for all $j \in J$ and $\lambda > 0$, we have $y_j(x_0 + \lambda v) \le y_j(x_0) = b_j$. Hence for small enough $\lambda > 0$, we have $x_0 + \lambda v \in P$. Thus $v \in D(P, x_0)$, as desired. $\square$

Corollary 3.1.7. Every face of a polyhedron is a support set of the polyhedron.

Proof. By Proposition 3.1.6, for any face F we have that $D(P, F)$ is polyhedral and therefore closed. Thus $D(P, F)$ has balanced support by Theorem 2.3.5. $\square$

Corollary 3.1.8. Let P, F, and J be as in Proposition 3.1.6. Then

$$F = P \cap \{y_j(x) = b_j,\ j \in J\}.$$

Proof. This follows from Propositions 3.1.1 and 2.1.9. $\square$

Corollary 3.1.9. A polyhedron has finitely many faces.

Proof. In Corollary 3.1.8, there are finitely many possible J. $\square$

3.2 Face posets

For a convex set K, let $\mathscr{F}(K)$ denote the set of all faces of K. We view $\mathscr{F}(K)$ as a poset ordered by inclusion. We call $\mathscr{F}(K)$ the *face poset* of K.

Proposition 3.2.1. Let K be a convex set. The following are true.

(a) Let F be a face of K and let G be any subset of F. Then G is a face of F if and only if G is a face of K.

(b) If $\{F_i : i \in I\}$ is a collection of faces of K and $\bigcap_{i \in I} F_i \neq \varnothing$, then $\bigcap_{i \in I} F_i$ is a face of K.

(c) If $K = K' \oplus L$ where L is a vector space, then the map $F \mapsto F \oplus L$ is a poset isomorphism $\mathscr{F}(K') \to \mathscr{F}(K)$.

(d) If K is nonempty, then K is the unique maximal element of $\mathscr{F}(K)$.

(e) If K is a cone, then $\operatorname{lins} K$ is the unique minimal element of $\mathscr{F}(K)$.

(f) If K is nonempty and closed, the minimal elements of $\mathscr{F}(K)$ are translates of $\operatorname{lins} K$.

Proof. Parts (a)–(e) are straightforward.

Proof of (f). By (c), we may assume $\operatorname{lins} K = \{0\}$. By Corollary 3.1.3, every face of K is closed. Then by Theorem 2.5.2, every face of K has an extreme point, and by (a), these are also extreme points of K. Hence the minimal elements of $\mathscr{F}(K)$ are the extreme points of K. $\square$

Proposition 3.2.2. Let K, K' be convex sets and $f : K \to K'$ an affine isomorphism. Then the map $F \mapsto f(F)$ is a poset isomorphism $\mathscr{F}(K) \to \mathscr{F}(K')$.

Proof. Easy from the definition of a face. □

Proposition 3.2.3. Let K be a convex set and F a face of K.

(a) The poset $\mathscr{F}(K)_{\le F} := \{G \in \mathscr{F}(K) : G \subset F\}$ is the face poset of F.

(b) The poset $\mathscr{F}(K)_{\ge F} := \{G \in \mathscr{F}(K) : G \supset F\}$ is isomorphic to the face poset of $D(K, F)$.

Proof. Part (a) is Proposition 3.2.1(a). For part (b), consider the map $\mathscr{F}(K)_{\ge F} \to \mathscr{F}(D(K, F))$ given by $G \mapsto D(G, F)$. We leave it as an exercise to show that this map is well-defined and an isomorphism (Exercise 3.6). □

Grading

Let $(\mathscr{P}, \le)$ be a poset. For x, $y \in \mathscr{P}$, we say that x *covers* y if $x > y$ and there is no z such that $x > z > y$. A *grading* on $\mathscr{P}$ is a function $\rho : \mathscr{P} \to \mathbb{Z}$ such that $\rho(x) > \rho(y)$ if $x > y$, and $\rho(x) = \rho(y) + 1$ if x covers y.

Proposition 3.2.4. For any polyhedron P, the dimension function is a grading on $\mathscr{F}(P)$.

Proof. Suppose F, $G \in \mathscr{F}(P)$ and G covers F. Then the interval $[F, G]$ in $\mathscr{F}(P)$ has only two elements. By Proposition 3.2.3, $[F, G]$ is isomorphic to $\mathscr{F}(D(G, F))$. Write $D(G, F) = C \oplus \operatorname{lins} D(G, F)$, where C is a polyhedral cone with $\operatorname{lins} C = \{0\}$. By Proposition 3.2.1(c), we have $\mathscr{F}(C) \cong \mathscr{F}(D(G, F)) \cong [F, G]$. So $\mathscr{F}(C)$ has exactly two elements $\{0\}$ and C. Since C is polyhedral and thus closed, by Theorem 2.4.2 it has at least one extreme ray, and therefore C itself is a ray. Hence $\dim C = 1$. Thus, $\dim D(G, F) - \dim \operatorname{lins} D(G, F) = 1$. Since $\dim D(G, F) = \dim G$ and $\dim \operatorname{lins} D(G, F) = \dim F$, we have $\dim G - \dim F = 1$, as desired. □

Facets

Given a polyhedron P of dimension d, the $(d-1)$-dimensional faces of P are called *facets* of P. Given a facet F of P, an inequality $y(x) \le b$ such that $\operatorname{argmax}_P y = F$ and $y(F) = b$ is called a *facet-defining inequality* of P for F. The facet-defining inequalities satisfy a minimal presentation theorem analogous to Theorem 2.4.2. (In fact, the following theorem can be deduced from Theorem 2.4.2 using polarity, but we will give a separate proof.)

Theorem 3.2.5. Let $P \subset V$ be a polyhedron. The following are true.

(a) For each facet F of P, let $y_F(x) \leq b_F$ be a facet-defining inequality for F. Then

$$P = \{y_F(x) \leq b_F,\, F \text{ a facet}\} \cap \operatorname{aff} P.$$

(b) Assume

$$P = \{y_i(x) \leq b_i,\, i \in I\}$$

where I is finite. Then for every facet F of P there is some $i \in I$ such that $y_i(x) \leq b_i$ is a facet-defining inequality for F.

Proof. Proof of (a). Let $P' = \{y_F(x) \leq b_F,\, F \text{ a facet}\} \cap \operatorname{aff} P$. Clearly $P \subset P'$. Suppose $x_0 \in P' \setminus P$. Let $u \in \operatorname{rint} P$, and let $v \in [u, x_0] \cap \operatorname{rbd} P$. We know that v is contained in some proper face of P, so by Proposition 3.2.4 it is contained in some facet F of P. Since $u \in P \setminus F$, we have $y_F(u) < b_F$, and hence $y_F(x_0) > b_F$. This contradicts $x_0 \in P'$.

Proof of (b). Let F be a facet of P. By Proposition 3.1.8, $F = P \cap \{y_j(x) = b_j,\, j \in J\}$ for some $J \subset I$. Since F is $(d-1)$-dimensional, $\{y_j(x) = b_j,\, j \in J\}$ intersects $\operatorname{aff} P$ at a hyperplane H of $\operatorname{aff} P$. This implies there is some $j \in J$ such that $\{y_j(x) = b_j\} \cap \operatorname{aff} P = H$. Then $y_j(x) \leq b_j$ is a defining inequality for F, as desired. □

Corollary 3.2.6. Every proper face of a polyhedron is the intersection of the facets that contain it.

Proof. This follows from Theorem 3.2.5(a) and Corollary 3.1.8. □

3.3 Examples

Example 3.3.1 (Simplicial cones)**.** Let

$$\mathbb{R}^n_{\geq 0} = \{x \in \mathbb{R}^n : e^i(x) \geq 0 \text{ for all } i\}$$

be the nonnegative orthant of $\mathbb{R}^n$ from Example 1.2.3. The extreme rays are generated by the vectors e_i, $i \in [n]$. The facet-defining inequalities are $e^i(x) \geq 0$. The faces are all sets of the form

$$\mathbb{R}^S_{\geq 0} := \operatorname{cone}\{e_i : i \in S\} = \mathbb{R}^n_{\geq 0} \cap \{e^i(x) = 0,\, i \notin S\}$$

where S is any subset of $[n]$. The map $S \mapsto \mathbb{R}^S_{\geq 0}$ is a poset isomorphism from the *Boolean lattice* $\mathscr{B}_n$ of subsets of $[n]$ to $\mathscr{F}(\mathbb{R}^n_{\geq 0})$.

Example 3.3.2 (Simplices)**.** Let

$$\Delta = \{x \in \mathbb{R}^{n+1} : e^1(x) + \cdots + e^{n+1}(x) = 1,\ e^i(x) \geq 0,\ i = 1, \ldots, n\}$$

be the n-dimensional simplex from Example 1.2.8. The vertices are the points e_i, and the facet-defining inequalities are $e^i(x) \geq 0$. The faces are the sets of the form

$$\Delta_S := \operatorname{conv}\{e_i : i \in S\} = \Delta \cap \{e^i(x) = 0,\, i \notin S\}$$

where S is any nonempty subset of $[n+1]$. The map $S \mapsto \Delta_S$ is a poset isomorphism from $\mathscr{B}_{n+1} \smallsetminus \{\varnothing\}$ to $\mathscr{F}(\Delta)$.

Example 3.3.3 (Cube)**.** Let

$$Q = \{x \in \mathbb{R}^n : -1 \leq e^i(x) \leq 1,\, i = 1, \ldots, n\}$$

be the n-dimensional cube from Example 1.2.6. The vertices of Q are all points of the form $\pm e_1 \pm \cdots \pm e_n$. The facet-defining inequalities are $e^i(x) \geq -1$ and $e^i(x) \leq 1$ over all $i \in [n]$. The faces of Q are

$$Q_{S,T} := Q \cap \{e^i(x) = -1,\, i \in S\} \cap \{e^i(x) = 1,\, i \in T\}$$

where S, T are any disjoint subsets of $[n]$. There are 3^n faces in total.

Example 3.3.4 (Cross-polytope)**.** Let

$$R = \{x \in \mathbb{R}^n : \pm e^1(x) \pm \cdots \pm e^n(x) \leq 1\},$$

be the d-dimensional cross-polytope from Example 1.2.7. The vertices of R are the points $\pm e_1, \ldots, \pm e_n$, and the facet-defining inequalities are the inequalities in the above presentation. The faces of R are

$$R_{S,T} := \operatorname{conv}(\{e_i : i \in S\} \cup \{-e_i : i \in T\})$$

where S, T are any disjoint subsets of $[n]$ with $S \cup T \neq \varnothing$, as well as R itself. There are 3^n faces in total.

3.4 Normal fans

In this section we construct the normal fan of a polyhedron, a structure which partitions the space of linear functionals according to the which face of the polyhedron maximizes that linear functional. The importance will be more apparent in later chapters.

Normal cones

Let $K \subset V$ be a convex set and $x \in K$. The *(outward) normal cone* of K at x is defined to be

$$N(K, x) := D(K, x)^{\circ}.$$

Recall that if F is a face of K and $x, x' \in \operatorname{rint} F$, then $D(K, x) = D(K, x')$. Thus we can define $N(K, F)$ to be $N(K, x)$ for any $x \in \operatorname{rint} F$. Elements of $N(K, x)$ or $N(K, F)$ are called *normal vectors* to K at x or F.

Proposition 3.4.1. Let K be a convex set, F a face of K, and $x \in \operatorname{rint} F$. The following are true.

(a) We have $y \in N(K, F)$ if and only if $F \subset \operatorname{argmax}_K y$, if and only if $x \in \operatorname{argmax}_K y$.

(b) If $D(K, F)$ is closed, then $y \in \operatorname{rint} N(K, F)$ if and only if $F = \operatorname{argmax}_K y$.

Proof. Proof of (a). The first part is Proposition 3.1.5(a), and then the second part follows from Exercise 2.6.

Proof of (b). Follows from Proposition 3.1.5(b) and Proposition 2.3.5. □

Corollary 3.4.2. Let K be a convex set and let F and G be faces of K with $F \subset G$. Then $N(K, G)$ is a support set of $N(K, F)$.

Proof. Let $x_F \in \operatorname{rint} F$ and $x_G \in \operatorname{rint} G$. By Proposition 3.4.1(a), for any $y \in N(K, F)$ we have $y(x_F) \geq y(x_G)$, and moreover $y \in N(K, G)$ if and only if $y(x_F) = y(x_G)$. Hence $N(K, G) = \operatorname{argmax}_{N(K,F)}(x_G - x_F)$, so $N(K, G)$ is a support set of $N(K, F)$. □

Proposition 3.4.3. If $D(K, F)$ is closed, then

$$\operatorname{span} N(K, F) = F^{\perp}.$$

Proof. This follows from Propositions 2.3.3 and 3.1.1(b). □

Example 3.4.4. Let $P = \{y_i(x) \leq b_i,\ i \in I\}$ be a polyhedron, where I is finite. By Proposition 3.1.6, for any face F of P we have $D(P, F) = \{y_j(x) \leq 0,\ j \in J\}$ where J is the set of all j such that $y_j(x) = b_j$ for all $x \in F$. Then by Proposition 2.3.2,

$$N(P, F) = \operatorname{cone}(\{y_j : j \in J\}).$$

Suppose instead that P is presented as $\{y_i(x) \leq b_i,\, i \in I\} \cap A$, where I is finite and A is an affine space. Then for any face F of P, we have $D(P, F) = \{y_j(x) \leq 0,\, j \in J\} \cap A$ where J is as before, and by Exercise 2.16,

$$N(P, F) = \operatorname{cone}(\{y_j : j \in J\}) + A^{\perp}.$$

Normal fans

For a convex set K, we define the *normal fan* $\mathscr{N}(K)$ of K to be the set of all normal cones of K, that is

$$\mathscr{N}(K) = \{N(K, x) : x \in K\} = \{N(K, F) : F \in \mathscr{F}(K)\}.$$

We make $\mathscr{N}(K)$ a poset by partially ordering by inclusion. We will focus on the case where K is a polyhedron.

Theorem 3.4.5. Let $P \subset V$ be a polyhedron. The following are true.

(a) The map $F \mapsto N(P, F)$ is a poset anti-isomorphism $\mathscr{F}(P) \to \mathscr{N}(P)$ with $\dim F + \dim N(P, F) = \dim V$.

(b) If P is nonempty, then $\mathscr{N}(P)$ has unique minimal element $N(P, P) = P^{\perp}$. In particular, the minimal element is $\{0\}$ if P is full-dimensional.

(c) For any face F of P, the faces of $N(P, F)$ are precisely the sets $N(P, G)$ where G is a face of P containing F.

(d) For nonempty P, we have $\bigcup_{F \in \mathscr{F}(P)} N(P, F) = (\operatorname{recc} P)^{\circ}$.

Proof. Proof of (a). The map is order-reversing by Corollary 3.4.2, and it is surjective by definition. Since P is a polyhedron, $D(P, F)$ is closed for all F. Thus the injectivity follows from Proposition 3.4.1(b). The equality $\dim F + \dim N(P, F) = \dim V$ follows from Proposition 3.4.3.

Proof of (b). From (a), $\mathscr{N}(P)$ has unique minimal element $N(P, P) = D(P, P)^{\circ}$. Since $D(P, P)$ is the linear subspace parallel to $\operatorname{aff} P$, we have $D(P, P)^{\circ} = D(P, P)^{\perp} = P^{\perp}$.

Proof of (c). By Proposition 3.4.1(a), for every $y \in N(P, F)$, $\operatorname{argmax}_P y$ is a face of P containing F. Hence, by Proposition 3.4.1(b), every $y \in N(P, F)$ is contained in $\operatorname{rint} N(P, G)$ for some face $G \supset F$ of K. By Corollary 3.4.2 each $N(P, G)$ is a face of $N(P, F)$, so by Corollary 3.1.2 these must be all the faces of $N(P, F)$.

Proof of (d). By Proposition 3.4.1(b), a linear functional is in $\bigcup_{F \in \mathscr{F}(P)} N(P, F)$ if and only if it achieves a maximum on P. The result then follows from Theorem 2.5.7. $\square$

Example 3.4.6 (Polyhedral cones)**.** If C is a polyhedral cone, then it has a unique minimal face $\operatorname{lins} C$. So by Theorem 3.4.5(a), $\mathscr{N}(C)$ has a unique maximal element $N(C, \{0\}) = D(C, \{0\})^\circ = C^\circ$. By Theorem 3.4.5(c), the other normal cones $N(C, F)$ are exactly the faces of C°. We thus obtain the following.

Proposition 3.4.7. If C is a polyhedral cone, then $\mathscr{N}(C) = \mathscr{F}(C^\circ)$. The map $F \mapsto N(C, F)$ is a poset anti-isomorphism $\mathscr{F}(C) \to \mathscr{F}(C^\circ)$.

Example 3.4.8 (Simplex)**.** Let Δ be the n-dimensional simplex from Example 1.2.8. The minimal element L of $\mathscr{N}(\Delta)$ is

$$\Delta^\perp = \{e^1(x) + \cdots + e^{n+1}(x) = 0\}^\perp = \operatorname{span}\{e^1 + \cdots + e^{n+1}\}.$$

The facet-defining inequalities are $-e^i(x) \leq 0$. Therefore, by Example 3.3.2, the next-to-minimal elements of $\mathscr{N}(\Delta)$ are $\operatorname{cone}\{-e_i\} + L$. From Example 3.3.2, the normal cone of the face Δ_S is $\operatorname{cone}\{-e_i : i \notin S\} + L$.

Exercises

3.1. Every face of a closed convex set is closed.
3.2. Let K, L be convex sets. The faces of $K \times L = \{(k, l) : k \in K, l \in L\}$ are precisely the sets $F \times G$ where F is a face of K and G is a face of L.
3.3. Let K, L be convex sets in the same space. The faces of $K \cap L$ are precisely the nonempty sets of the form $F \cap G$ where F is a face of K and G is a face of L.
3.4. Let K be a convex set. If x, $x' \in \operatorname{rint} F$ for some face F of K, then $D(K, x) = D(K, x')$.
3.5. (a) Let C be a closed convex set with $\operatorname{lins} C = \{0\}$ and F a face of C. Then F is the conical hull of all extreme rays of C contained in F.
(b) Let K be a closed bounded convex set and F a face of K. Then F is the convex hull of all extreme points of K contained in F.
3.6. Let K be a convex set and F a face of K. The map $G \mapsto D(G, F)$ is a poset isomorphism $\mathscr{F}(K)_{\geq F} \to \mathscr{F}(D(K, F))$.
3.7. Let K be a convex set and $\operatorname{homog} K$ its homogenization. Then the map $F \mapsto \operatorname{homog} F$ is an injective map of posets $\mathscr{F}(K) \to \mathscr{F}(\operatorname{homog} K)$. The faces of $\operatorname{homog} K$ not in the image of this map are $F \times \{0\}$, where F is a face of $\operatorname{recc} K$.

3.8. Let K be the following convex subset of $\mathbb{R}^2$:

$$K = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \le 1\} \cup \{0 \le x \le 1,\ 0 \le y \le 1\}.$$

Then $(0, 1)$ and $(1, 0)$ are faces of K but not support sets of K.

3.9. Let K be a convex set and F a face of K. Let $y \in V^*$. The following are true.

(a) $F \subset \operatorname{argmax}_K y$ if and only if $y \in D(K, F)^\circ$.

(b) $F = \operatorname{argmax}_K y$ if and only if $\operatorname{argmax}_{D(K,F)} y = \operatorname{lins} D(K, F)$.

3.10. If P is a polyhedral cone with $\dim P = \dim V$, then an inequality $y(x) \le 0$ is facet-defining if and only if y is an extreme ray of P°.

3.11. Let K be d-dimensional convex set and F a face of dimension $d - 1$. Then F is a support set of K.

Chapter 4

Convex functions

4.1 Convex functions

We will work in the *extended real number system* $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$. Arithmetic in $\overline{\mathbb{R}}$ is defined in the "obvious" way; for example, $a + \infty = \infty$ and $a \cdot \infty = \infty$ for all $a \in \mathbb{R} \cup \{\infty\}$. Expressions such as $-\infty + \infty$ and $0 \cdot \infty$ are undefined.

Let V be a real finite-dimensional vector space. Let $f : V \to \overline{\mathbb{R}}$ be any function. The *epigraph* of f is the set $\operatorname{epi} f \subset V \times \mathbb{R}$ defined by

$$\operatorname{epi} f := \{(x, s) \in V \times \mathbb{R} : f(x) \leq s\}.$$

A function $f : V \to \overline{\mathbb{R}}$ is *convex* if $\operatorname{epi} f$ is a convex set. Equivalently, f is convex if and only if

$$f((1-\lambda)x_1 + \lambda x_2) \leq (1-\lambda)f(x_1) + \lambda f(x_2) \tag{4.1.1}$$

for all x_1, $x_2 \in V$ such that $f(x_1)$, $f(x_2) < \infty$ and all $0 < \lambda < 1$.

Example 4.1.2. Given a set $K \subset V$, define the *indicator function* of K to be the function $\iota_K : V \to \mathbb{R}$ given by

$$\iota_K(x) = \begin{cases} 0 & \text{if } x \in K \\ \infty & \text{if } x \notin K. \end{cases}$$

We have that K is a convex set if and only if ι_K is a convex function.

For a function $f : V \to \overline{\mathbb{R}}$, the *effective domain* $\operatorname{dom} f$ is the set of all $x \in V$ for which $f(x) < \infty$. If f is a convex function, then $\operatorname{dom} f$ is a convex set.

The constant functions $f(x) = -\infty$ and $f(x) = \infty$ are both convex; we denote them by $V^{-\infty}$ and V^{∞}, respectively.

Operations on functions

Let S be a subset of $\overline{\mathbb{R}}$. We define the *supremum* $\sup S$ and *infimum* $\inf S$ of S the usual way if S is nonempty. If S is empty, then we define $\sup S = -\infty$ and $\inf S = \infty$.

Let $\{f_i\}_{i\in I}$ be a set of functions $f_i : V \to \overline{\mathbb{R}}$. We define $\sup\{f_i\} : V \to \overline{\mathbb{R}}$ to be the pointwise supremum of $\{f_i\}_{i\in I}$, that is

$$(\sup\{f_i\})\,(x) = \sup_{i\in I} f_i(x) \quad \text{for all } x \in V.$$

We similarly define $\inf\{f_i\} : V \to \overline{\mathbb{R}}$ to be the pointwise infimum of $\{f_i\}$.

Proposition 4.1.3. The following are true.

(a) If $\{f_i\}_{i\in I}$ is a set of functions $V \to \overline{\mathbb{R}}$, then $\operatorname{epi}\sup\{f_i\} = \bigcap_{i\in I} \operatorname{epi} f_i$. In particular, if all the f_i are convex, then $\sup\{f_i\}$ is convex.

(b) If f, $g : V \to \mathbb{R} \cup \{\infty\}$ are convex functions, then $f + g$ is a convex function.

Proof. Exercise 4.3. □

For any function $f : V \to \overline{\mathbb{R}}$, define the *convex envelope* of f to be the function

$$\operatorname{env} f := \sup\{g : V \to \overline{\mathbb{R}} : g \text{ is convex and } g \leq f\}.$$

Here, $g \leq f$ means that $g(x) \leq f(x)$ for all $x \in V$. By Proposition 4.1.3(a), $\operatorname{env} f$ is a convex function, and $f = \operatorname{env} f$ if and only if f is convex.

Subgradients

Let $f : V \to \overline{\mathbb{R}}$ be a convex function, and let $x_0 \in V$. We say a linear functional $y \in V^*$ is a *subgradient* of f at x_0 if

$$f(x) \geq y(x - x_0) + f(x_0) \tag{4.1.4}$$

for all $x \in V$. If $f(x_0)$ is finite, then y is a subgradient of f at x_0 if and only if the graph of the affine function $y(x - x_0) + f(x_0)$ is tangent to the graph of f at $(x_0, f(x_0))$, or equivalently, if the vector $(y, -1)$ is a normal vector of $\operatorname{epi} f$ at $(x_0, f(x_0))$. If f is differentiable at x_0, then it has a unique subgradient at x_0 equal to the usual gradient $\nabla f(x_0)$.

The set of all subgradients of f at x_0 is called the *subdifferential* of f at x_0 and is denoted $\partial f(x_0)$. The set $\partial f(x_0)$ is a closed convex subset of V^* (Exercise 4.4).

Proposition 4.1.5. Let $f : V \to \overline{\mathbb{R}}$ be a convex function and $x_0 \in \operatorname{rint} \operatorname{dom} f$. Then $\partial f(x_0)$ is nonempty.

Proof. If $f(x_0) = -\infty$ then $\partial f(x_0) = V^*$ and we are done. Thus assume $f(x_0)$ is finite. Then $(x_0, f(x_0)) \in \operatorname{bd} \operatorname{epi} f$. By the supporting hyperplane theorem applied to $\operatorname{epi} f$, there exists $(y, t) \in V^* \times \mathbb{R}^*$ such that $(y, t)(x, s) \leq (y, t)(x_0, f(x_0))$ for all $(x, s) \in \operatorname{epi} f$, and (y, t) is not constant on $\operatorname{epi} f$. If $t < 0$, then by scaling we may assume $t = -1$. Then $(y, -1)$ is a normal vector of $\operatorname{epi} f$ at $(x_0, f(x_0))$, so $y \in \partial f(x_0)$ as desired.

If $t > 0$, then for $s > f(x_0)$ we have $(y, t)(x_0, s) > (y, t)(x_0, f(x_0))$, a contradiction since $(x_0, s) \in \operatorname{epi} f$. If $t = 0$, then $y(x) \leq y(x_0)$ for all $x \in \operatorname{dom} f$. Since $x_0 \in \operatorname{rint} \operatorname{dom} f$, this implies y is constant on $\operatorname{dom} f$, in which case (y, t) is constant on $\operatorname{epi} f$, a contradiction. □

4.2 Convex conjugation

Closed convex functions

Let $f : V \to \overline{\mathbb{R}}$ be a convex function. We define the *closure* of f to be the function $\operatorname{cl} f : V \to \overline{\mathbb{R}}$ given as follows.

(a) If $f(x) = -\infty$ for any $x \in V$, then $\operatorname{cl} f = V^{-\infty}$.

(b) Otherwise, $\operatorname{cl} f$ is the unique function $V \to \overline{\mathbb{R}}$ whose epigraph is $\operatorname{cl} \operatorname{epi} f$.

A convex function f is *closed* if $f = \operatorname{cl} f$. Thus, f is closed if and only if $\operatorname{epi} f$ is closed, and either $f = V^{-\infty}$ or $f(V) \subset \mathbb{R} \cup \{\infty\}$.

The epigraphs of closed convex functions can be separated from outside points by "non-vertical hyperplanes". This is stated precisely below.

Proposition 4.2.1. Let $f : V \to \overline{\mathbb{R}}$ be a closed convex function and let $(x_0, s_0) \in V \times \mathbb{R} \setminus \operatorname{epi} f$. Then there exists $y \in V^*$ such that for all $x \in V$,

$$y(x) - f(x) < y(x_0) - s_0.$$

Proof. The statement is easy if $f = V^{-\infty}$ or V^{∞}, so we may assume otherwise. Since $\operatorname{epi} f$ is closed, by the separating hyperplane theorem, there exists $(y, t) \in V^* \times \mathbb{R}^*$ such that $\max_{\operatorname{epi} f}(y, t) < (y, t)(x_0, s_0)$. If $t < 0$, then we may assume assume $t = -1$ and we are done. Suppose $t > 0$. Since $\operatorname{epi} f$ is nonempty, there exists $(x, s) \in \operatorname{epi} f$. Then $(x, s') \in \operatorname{epi} f$ for all $s' > s$, so (y, t) is unbounded from above on $\operatorname{epi} f$, a contradiction.

Finally, assume $t = 0$. Then $\max_{\operatorname{dom} f} y < y(x_0)$. Hence there is some $\varepsilon > 0$ such that $y(x_0 - x) > \varepsilon$ for all $x \in \operatorname{dom} f$. Let x_1 be any point in $\operatorname{rint} \operatorname{dom} f$. By Proposition 4.1.5, there exists $y_1 \in V^*$ such that

$$(y_1, -1)(x, s) \leq (y_1, -1)(x_1, f(x_1))$$

for all $(x, s) \in \operatorname{epi} f$. Let $y' = \lambda y + y_1$ for some $\lambda > 0$. Then

$$(y', -1)((x_0, s_0) - (x, f(x))) \geq \lambda\varepsilon + (y_1, -1)((x_0, s_0) - (x_1, f(x_1)))$$

for all $(x, f(x)) \in \operatorname{epi} f$. Since f is closed and not $V^{-\infty}$, $f(x_1)$ is finite. So we can choose λ large enough so the right hand side is positive, completing the proof. □

Proposition 4.2.2. The following are true.

(a) If $\{f_i\}_{i \in I}$ is a set of closed convex functions $V \to \overline{\mathbb{R}}$, then $\sup\{f_i\}$ is a closed convex function.

(b) If $f,\ g : V \to \mathbb{R} \cup \{\infty\}$ are closed convex functions, then $f + g$ is a closed convex function.

Proof. Exercise 4.5. □

Convex conjugates

Let $f : V \to \overline{\mathbb{R}}$ be any function. We define the *convex conjugate* of f to be the function $f^* : V^* \to \overline{\mathbb{R}}$ given by

$$f^*(y) = \sup_{x \in V}\{y(x) - f(x)\}.$$

For a fixed $x \in V$, the expression $y(x) - f(x)$ is either an affine function of y, or equals ∞ for all y or $-\infty$ for all y. In every case, $y(x) - f(x)$ is a convex and closed function of y. Thus f^* is convex and closed by Proposition 4.2.2.

One way to understand f^* is as follows. Let $y \in V^*$, and suppose y is a subgradient of f at some $x_0 \in V$. Then $y(x) - f(x) \leq y(x_0) - f(x_0)$ for all $x \in V$. Hence, by the definition of f^*, we have

$$f^*(y) = y(x_0) - f(x_0). \tag{4.2.3}$$

If $f(x_0)$ is finite, then the graph of the affine function $y - f^*(y)$ is tangent to the graph of f at $(x_0, f(x_0))$.

Theorem 4.2.4. Let $f : V \to \overline{\mathbb{R}}$ be a function. Then $f^{**} = \operatorname{cl}\operatorname{env} f$. In particular, if f is convex and closed, then $f^{**} = f$.

Proof. For any $x_0 \in V$, we have

$$\begin{aligned} f^{**}(x_0) &= \sup_{y \in V^*} \{y(x_0) - f^*(y)\} \\ &= \sup_{y \in V^*} \{y(x_0) - \sup_{x \in V}\{y(x) - f(x)\}\} \\ &\leq \sup_{y \in V^*} \{y(x_0) - (y(x_0) - f(x_0))\} \\ &= f(x_0). \end{aligned}$$

Thus $f^{**} \leq f$. Since f^{**} is convex and closed, we have $f^{**} \leq \operatorname{cl}\operatorname{env} f$. Thus it suffices to show $f^{**} \geq \operatorname{cl}\operatorname{env} f$.

Let $x_0 \in V$ and let $g = \operatorname{cl}\operatorname{env} f$. Suppose $\alpha < g(x_0)$. Then $(x_0, \alpha) \notin \operatorname{epi} g$. By Proposition 4.2.1, there exists $y \in V^*$ such that $y(x) - g(x) < y(x_0) - \alpha$ for all $x \in V$. Since $f \geq g$, we have $y(x) - f(x) < y(x_0) - \alpha$ for all $x \in V$. Hence $f^*(y) \leq y(x_0) - \alpha$ by definition of f^*. Thus $y(x_0) - f^*(y) \geq \alpha$, so by definition of f^{**}, we have $f^{**}(x_0) \geq \alpha$. This holds for all $\alpha < g(x_0)$, so $f^{**}(x_0) \geq g(x_0)$, as desired. □

Example 4.2.5. Let $f = y_0 - b$ be an affine function, where $y_0 \in V^*$ and $b \in \mathbb{R}$. Then

$$f^*(y) = \begin{cases} b & \text{if } y = y_0 \\ \infty & \text{otherwise.} \end{cases}$$

4.3 Support functions

Let $K \subset V$ be a set. The *support function* of K is the function $h_K : V^* \to \overline{\mathbb{R}}$ defined by

$$h_K(y) = \sup_{x \in K} y(x).$$

For a fixed $x \in K$, the expression $y(x)$ is a linear function of y. Thus h_K is convex and closed by Proposition 4.2.2.

Support functions are *nonnegative homogeneous*, that is, $h_K(\lambda y) = \lambda h_K(y)$ for all $\lambda \geq 0$ and $y \in V^*$. Along with convexity, this implies support functions are *sublinear* (Exercise 4.6). Conversely, every function $V \to \overline{\mathbb{R}}$ that is closed, convex, and nonnegative homogeneous is the support function of a closed convex set (Exercise 4.7).

Recall the definition of an indicator function from Example 4.1.2. We have the following fundamental relationship between indicator functions and support functions.

Proposition 4.3.1. For any set $K \subset V$, we have $h_K = \iota_K^*$. If K is convex and closed, then $h_K^* = \iota_K$.

Proof. We have

$$\iota_K^*(y) = \sup_{x \in V} \{y(x) - \iota_K(x)\}.$$

Since $y(x) - \iota_K(x)$ equals $y(x)$ if $x \in K$ and $-\infty$ otherwise, we have $\iota_K^*(y) = \sup_K y$ as desired. If K is convex and closed, then ι_K is convex and closed, so $h_K^* = \iota_K$ by Theorem 4.2.4. □

Recall that for two sets K, $L \subset V$, their Minkowsi sum is the set $K + L = \{k + l : k \in K, \, l \in L\}$.

Proposition 4.3.2. Let K, L be nonempty sets. Then $h_{K+L} = h_K + h_L$.

Proof. It is clear that for nonempty sets A, $B \subset \mathbb{R}$, we have $\sup(A + B) = \sup A + \sup B$. Thus for all $y \in V^*$,

$$h_{K+L}(y) = \sup_{\substack{x_1 \in K \\ x_2 \in L}} \{y(x_1) + y(x_2)\} = \sup_{x_1 \in K} y(x_1) + \sup_{x_2 \in L} y(x_2).$$

□

Proposition 4.3.3. Let $f : V \to \overline{\mathbb{R}}$. Then for all $y \in V^*$,

$$f^*(y) = h_{\operatorname{epi} f}(y, -1).$$

Proof. This follows from definitions. $\square$

Exercises

4.1. If $f : V \to \mathbb{R} \cup \{\infty\}$ is convex, then f is continuous at any $x \in \operatorname{rint} \operatorname{dom} f$.

4.2. A function $f : V \to \overline{\mathbb{R}}$ is *lower semi-continuous* if for all $x_0 \in V$ and $\alpha < f(x_0)$, there exists an open neighborhood U of x_0 such that $f(x) > \alpha$ for all $x_0 \in U$. Then a function $f : V \to \mathbb{R} \cup \{\infty\}$ is lower semi-continuous if and only if it is closed.

4.3. The following are true.

(a) If $\{f_i\}_{i \in I}$ is a set of functions $V \to \overline{\mathbb{R}}$, then $\operatorname{epi} \sup\{f_i\} = \bigcap_{i \in I} \operatorname{epi} f_i$. In particular, if all the f_i are convex, then $\sup\{f_i\}$ is convex.

(b) If $f,\ g : V \to \mathbb{R} \cup \{\infty\}$ are convex functions, then $f + g$ is a convex function.

4.4. The subdifferential $\partial f(x_0)$ is a closed convex subset of V^*.

4.5. The following are true.

(a) If $\{f_i\}_{i \in I}$ is a set of closed convex functions $V \to \overline{\mathbb{R}}$, then $\sup\{f_i\}$ is a closed convex function.

(b) If $f,\ g : V \to \mathbb{R} \cup \{\infty\}$ are closed convex functions, then $f + g$ is a closed convex function.

4.6. For all sets $K \subset V$, h_K is sublinear. That is, $h_K(\lambda y) = \lambda h_K(y)$ for all $\lambda \geq 0$ and $y \in V^*$, and $h_K(y_1 + y_2) \leq h_K(y_1) + h_K(y_2)$ for all $y_1,\ y_2 \in V^*$.

4.7. A function $V \to \overline{\mathbb{R}}$ is closed, convex, and nonnegative homogeneous if and only if it is the support function of a closed convex set. *Hint:* Use convex conjugation.